Akoizo Original Practice - Form 4 - Answer key

Division C | Hovercraft | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Which battery type is prohibited for powering the motors?

Unit: The electrical system: batteries, motors, and speed control.
Worked-case data: A team wants to use eight 1.5 V alkaline AA cells. Is that legal, and what are the options?
Separate investigation observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

A. Nickel-metal hydride

B. Nickel-cadmium

C. Lithium-ion

D. Alkaline

Correct: C

2. (2 points) A hovercraft drifts toward its left rail while hovering in place. A likely cause is:

A. Battery voltage too high

B. The dowel is too tall

C. More air leaking on the left side of the skirt

D. Too much nickel mass on the right

Correct: C

3. (2 points) Which action is allowed on the track during competition time?

Unit: Impound, inspection, and competition day.
Worked-case data: A team's vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

A. Dry-sweeping it

B. Wetting it

C. Asking the supervisor to re-level it

D. Applying a slick coating

Correct: A

4. (2 points) Where should the loaded center of mass sit?

Unit: Carrying nickels: mass, balance, and security.
Worked-case data: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.
Separate investigation observations:
Run 1: 4 rolls in cradles, 20 loose in an open tray: 3 nickels lost at start.
Run 2: Same load, tray with 2 cm walls: nothing lost.
Run 1 score effect: Penalty x 0.8 applied to the whole run.

A. Over the center of the air cushion

B. As far forward as possible

C. Directly over the thrust fan

D. Near one rail

Correct: A

5. (4 points) Explain how a fan produces forward thrust, using Newton's third law.

answer: The fan pushes air backward; the air pushes the fan, and so the vehicle, forward with an equal and opposite force. (4 points).

6. (4 points) Why can carrying more nickels lower your score even though Mass Score rises?

answer: More mass slows the vehicle and can make it stall or drop nickels, causing an incomplete run with TS = 0, a smaller DS, and a penalty. (4 points).

7. (4 points) Setting 6 gives 14.0 s and setting 8 gives 18.0 s. Estimate a setting for 15.0 s.

answer: (15 - 14) ÷ (18 - 14) = 0.25, so setting about 6 + 0.25 x 2 = 6.5. (4 points).

8. (4 points) State two safety rules a vehicle must pass to run.

Unit: The electrical system: batteries, motors, and speed control.
Worked-case data: A team wants to use eight 1.5 V alkaline AA cells. Is that legal, and what are the options?
Separate investigation observations:
Monday: Fresh pack 5.2 V, knob at 3: 11.8 s.
Wednesday: Same knob, pack at 4.7 V: 13.4 s.
Wednesday, recharged: Pack 5.2 V, knob at 3: 11.9 s.

answer: All propellers and impellers shielded against a 3/8-inch dowel; an accessible mechanical on/off switch for every motor; commercial non-lithium, non-lead batteries at 12.0 V or less. (4 points).

9. (4 points) How can the supervisor test whether a vehicle is levitating?

Unit: Levitation: the air cushion and the skirt.
Worked-case data: A hovercraft with nickels weighs 1.6 kg and has a 0.35 m x 0.35 m cushion. What cushion pressure is needed?
Separate investigation observations:
Empty test: Vehicle glides freely across the floor with lift on.
With all nickels: Rear skirt drags and the craft stops quickly.
Nickels moved forward 6 cm: Glides freely again, sitting level.

answer: By pressing the vehicle slightly downward; if it rises again on its own, it is supported by the air cushion. (4 points).

10. (4 points) How can the build log help with design-knowledge questions?

Unit: Impound, inspection, and competition day.
Worked-case data: A team’s vehicle passes inspection, but its first attempt stalls behind the start line for 4 seconds. What happens, and what should the team do?
Separate investigation observations:
Inspection: Rear impeller guard has a gap a 3/8-inch dowel fits through.
Team response: Covers the gap with spare mesh from impounded parts.
Re-check: Dowel can no longer reach the impeller.

answer: It records the design choices, test results, and calibration data that explain how and why the vehicle works. (4 points).

11. (6 points) Solve this case and explain your reasoning: A vehicle carries 3 rolls, the half roll, and 12 loose nickels, but 2 loose nickels fall out mid-run. The run is complete. Find the Mass Score and the penalty effect.

reasoning: Counted load: R = 3.5, N = 10, so 4 x 3.5 + 0.1 x 10 = 15.
Something fell off, so one construction penalty applies: DS, TS, and MS x 0.8.
MS after penalty = 15 x 0.8 = 12. (3 points).

conclusion: Two lost nickels cost 0.2 MS directly, but the penalty cost 20 percent of the whole run. (3 points).

12. (2 points) What two variables are changing the run time here?

Supplied synthetic observations:
Trial 1: Start 5 cm behind line: 14.1 s.
Trial 2: Start 40 cm behind line: 11.6 s.
Trial 3: Start 40 cm behind, plus 2 rolls: 13.8 s.

A. Only the timer

B. Start position and carried mass

C. Battery chemistry and skirt color

Correct: B

13. (2 points) Define “Target Time (TT)” in the context of Hovercraft.

answer: The run time every team aims for, set between 6.0-26.0 s. (2 points).

14. (2 points) Define “Calibration curve” in the context of Hovercraft.

answer: Graph of run time against a control setting. (2 points).

15. (2 points) Define “Parallel” in the context of Hovercraft.

answer: Cells side by side; same voltage, more capacity. (2 points).

16. (2 points) Define “Skirt” in the context of Hovercraft.

answer: Flexible curtain that contains the cushion air. (2 points).

17. (2 points) Define “Competition time” in the context of Hovercraft.

answer: 8 minutes for adjustments and two runs. (2 points).

18. (2 points) Define “Cradle” in the context of Hovercraft.

answer: Shaped holder that keeps a roll in place without tape. (2 points).